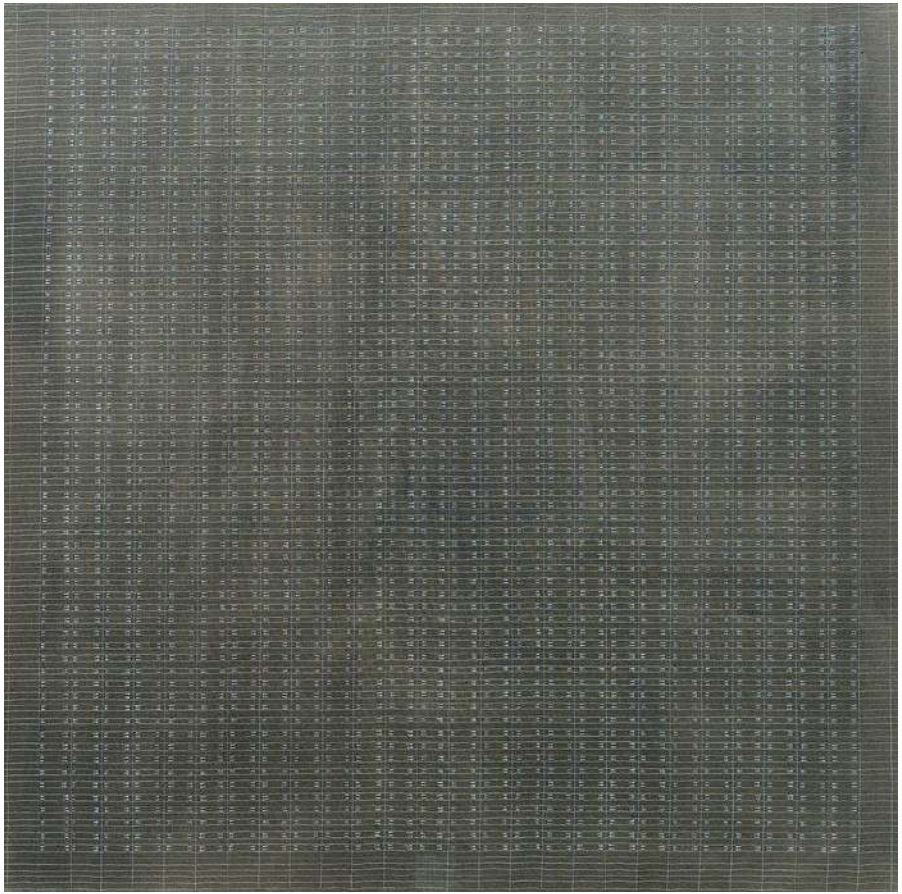
Why are AI systems biased?

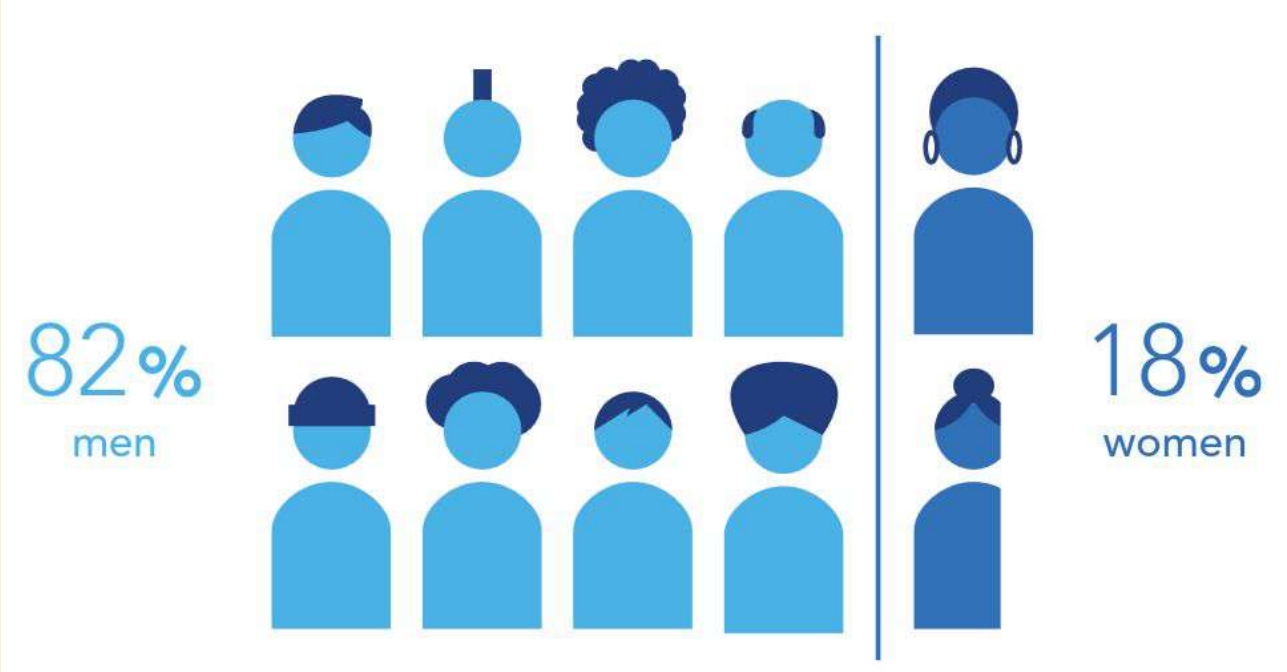


"White Flower" Agnes Martin, 1960

Images do not always describe themselves.

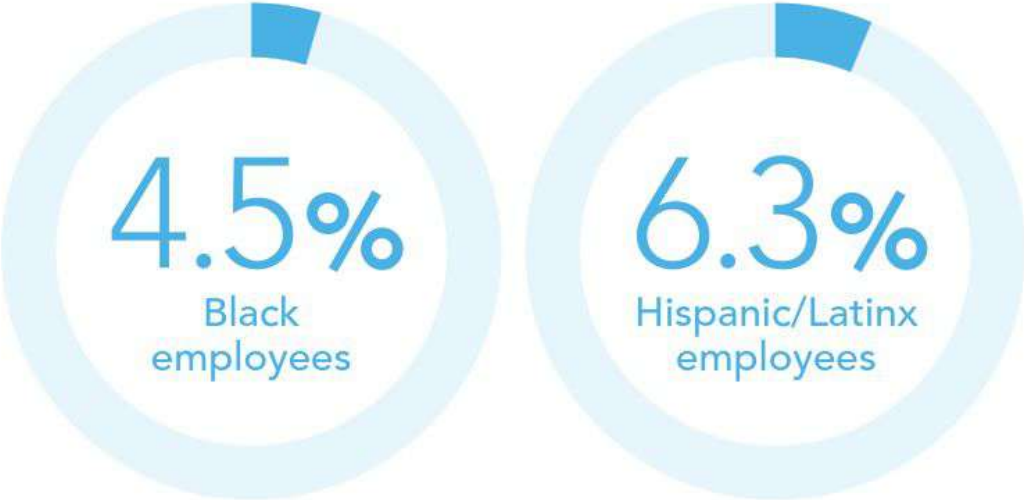
We see this image differently when we see how they are labeled.

It matters who develops AI systems.



Only 18% of researchers at leading AI conferences are women.

It matters who develops AI systems.



At Microsoft, 4.5% of employees are Black and 6.3% are Hispanic/Latinx. These numbers are similar at other tech firms.

"Artificial intelligence reflects the values, priorities, and prejudices of its creators." – Kate Crawford

Biases are being “baked in”

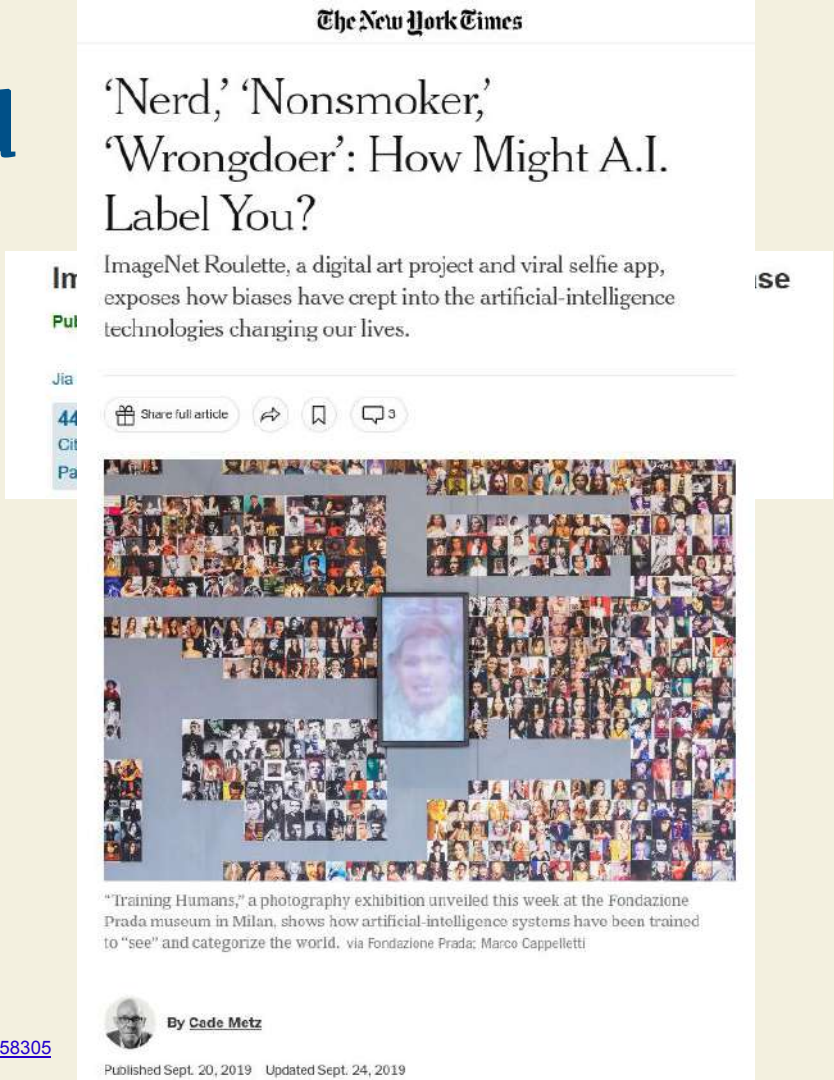
ImageNet, developed by researchers at Princeton and Stanford in 2009, is a widely used database with millions of images that computer vision AI technologies learn from.

Labeling: by thousands of unknown people, most likely in the United States, hired by the team from Stanford.

- Amazon Mechanical Turk,
- Earned money for each photo they labeled,
- hundreds of tags an hour.

J. Deng, W. Dong, R. Socher, L. -J. Li, Kai Li and Li Fei-Fei, "ImageNet: A large-scale hierarchical image database," 2009 IEEE Conference on Computer Vision and Pattern Recognition, Miami, FL, USA, 2009, pp. 248-255, doi: 10.1109/CVPR.2009.5206848.

https://news.artnet.com/art-world-archives/imagenet-roulette-trevor-paglen-kate-crawford-1658305



The New York Times

‘Nerd,’ ‘Nonsmoker,’ ‘Wrongdoer’: How Might A.I. Label You?

ImageNet Roulette, a digital art project and viral selfie app, exposes how biases have crept into the artificial-intelligence technologies changing our lives.

Share full article 3

“Training Humans,” a photography exhibition unveiled this week at the Fondazione Prada museum in Milan, shows how artificial-intelligence systems have been trained to “see” and categorize the world. via Fondazione Prada; Marco Cappelletti

By Cade Metz

Published Sept. 20, 2019 Updated Sept. 24, 2019

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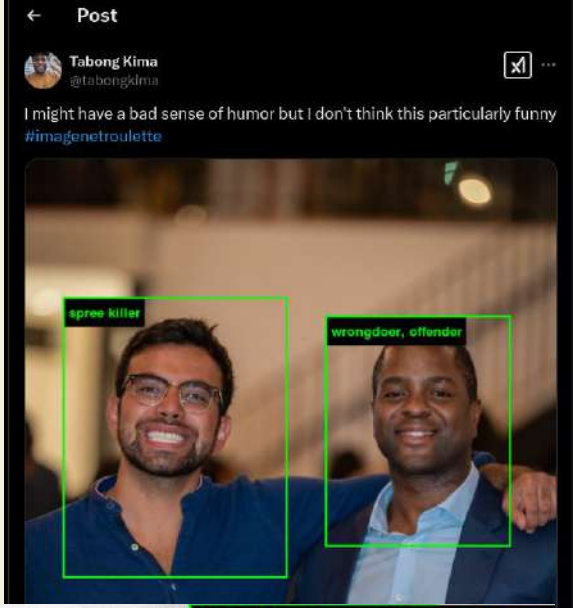

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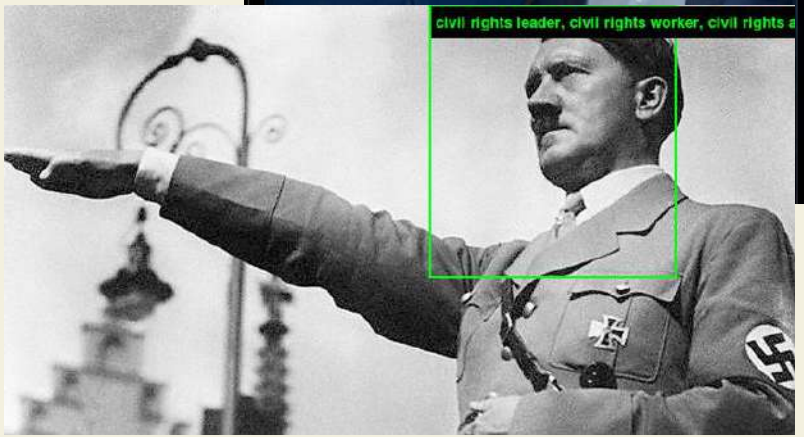
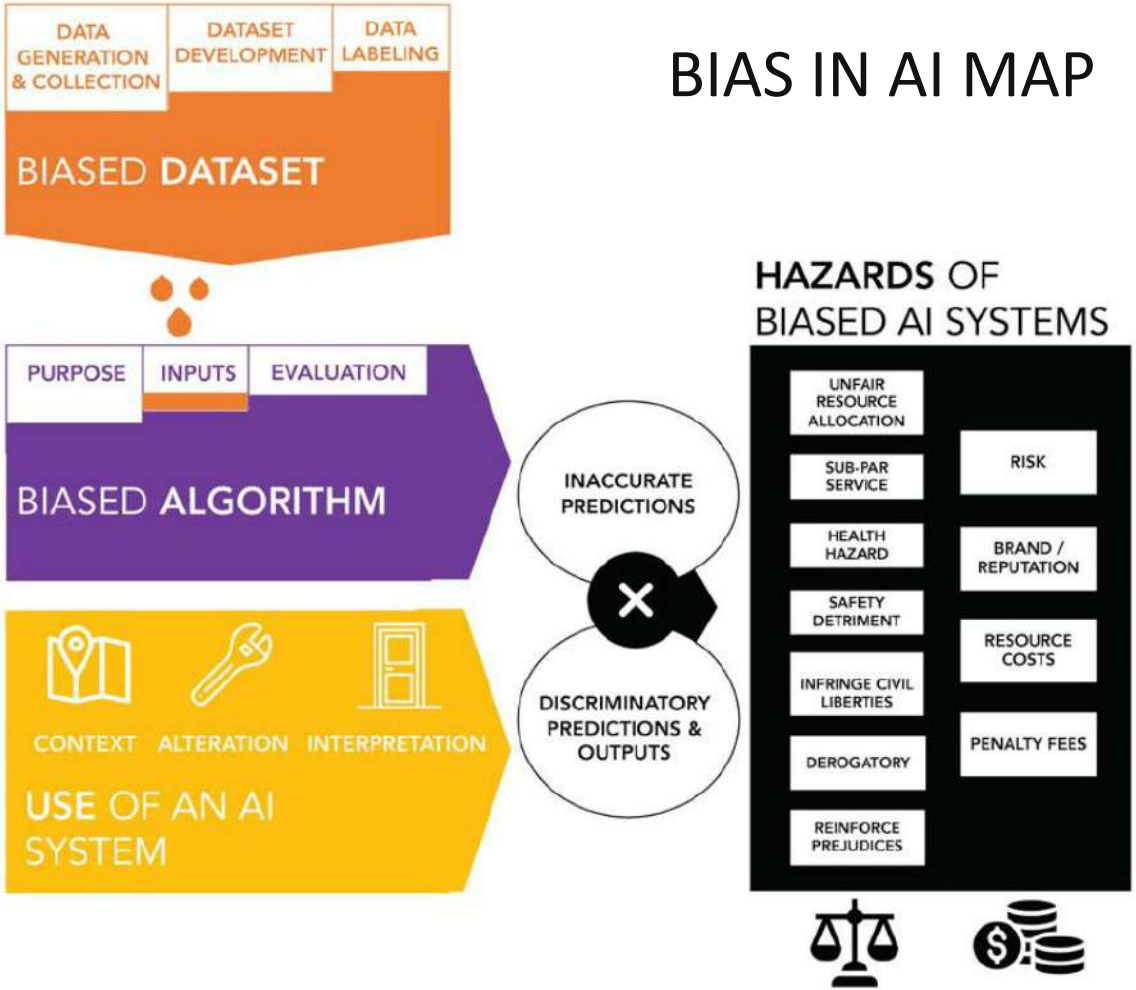


BIAS IN AI MAP



Mitigating Bias in Artificial Intelligence, Haas School of Business, University of California, Berkeley

Biased Dataset

- Data is assumed to accurately reflect the world, but there are significant data gaps (including **little or no data coming from particular communities**).
- **Human influence**: Data labeling can be Subjective.
- **Data collected** in the past preserves and reflects that past.
- Biased datasets can also be **accurate but representative of an unjust society.**

  In this case, they reflect biases against certain groups that are reflective of real discrimination that the particular group(s) face.

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arXiv:2403.20147v1 [cs.CL] 29 Mar 2024

IndiBias: A Benchmark Dataset to Measure Social Biases in Language Models for Indian Context

Nihar Ranjan Sahoo*, Pranamya Prashant Kulkarni*, Narjis Asad*, Arif Ahmad*,
Tanu Goyal♱, Aparna Garimella♰, Pushpak Bhattacharyya*
*Indian Institute of Technology Bombay, India
♱Google, India, ♰Adobe Research, India
{nihar, narjisasad, pb}@cse.iitb.ac.in, pranamyakulkarni@gmail.com
arifahmadpeace@gmail.com, tanugoyal@google.com, garimell@adobe.com

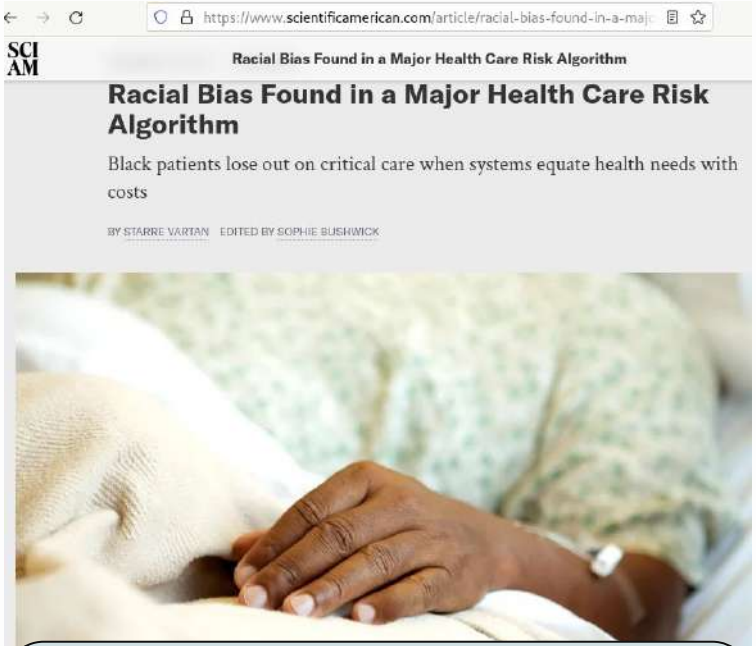
Data Collection

Data points may not exist for certain groups, identities or communities. This can be due to historical and current inequities.

As of 2016, >80% of all genetics data is from individuals of European ancestry.

Healthcare:
Men's bodies have always been the standard for medical testing. Women are missing from medical trials with female bodies deemed too complex and too variable.

The result? Approval of drugs with greater health risks for women and more misdiagnoses and fatalities of women from diseases.



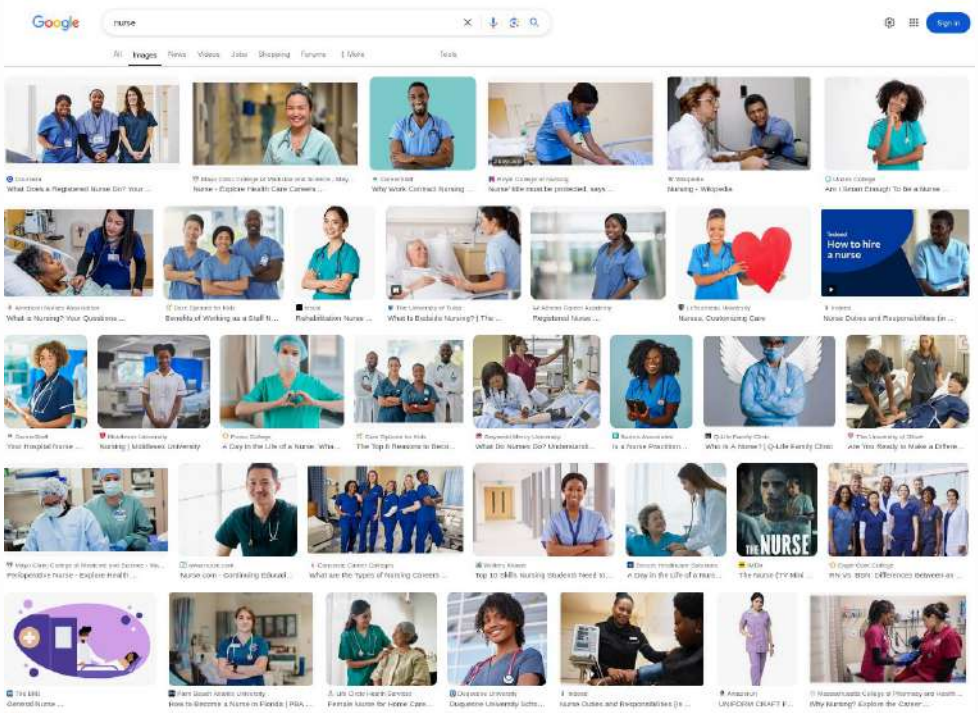
While nearly 40% of Americans identify as being non white, 80-90% of participants in most clinical trials are White.

Data Collection

When data points do exist, they may have prejudice built in and reflect inequities in society.

COMPAS: Racial Bias in Criminal Justice Systems

accurate but representative of an unjust society: Occupations such as “nurse” and “homemaker” are more likely to be assigned a female gender than occupations such as “doctor”.



Selection of data points to go into a dataset

Proceedings of Machine Learning Research 81:1–15, 2018 Conference on Fairness, Accountability, and Transparency

Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*

Joy Buolamwini JOYAB@MIT.EDU
MIT Media Lab 75 Amherst St. Cambridge, MA 02139
Timnit Gebru TIMNIT.GEBRU@MICROSOFT.COM
Microsoft Research 641 Avenue of the Americas, New York, NY 10011

Editors: Sorelle A. Friedler and Christo Wilson

Joy Buolamwini and Timnit Gebru highlighted the **lack of diverse and representative samples in image datasets** used for commercial facial recognition systems.

MIT News
ON CAMPUS AND AROUND THE WORLD

SUBSC

Study finds gender and skin-type bias in commercial artificial-intelligence systems

Examination of facial-analysis software shows error rate of 0.8 percent for light-skinned men, 34.7 percent for dark-skinned women.

FRS	Original	Blurry
AWS	Male	Male
Face++	Male	Male
Microsoft	Male	Male

No change in predicted gender

FRS	Original	Blurry
AWS	Female	Male
Face++	Female	Male
Microsoft	Male	Female

The predicted gender changed for all FRSs

https://www.media.mit.edu/publications/full-gender-shades-thesis-17/

Data Labeling

A **Beauty AI competition** trained robots on pre-labeled images of what was ‘beautiful’. The robots did not like people with dark skin.

Data Labeling

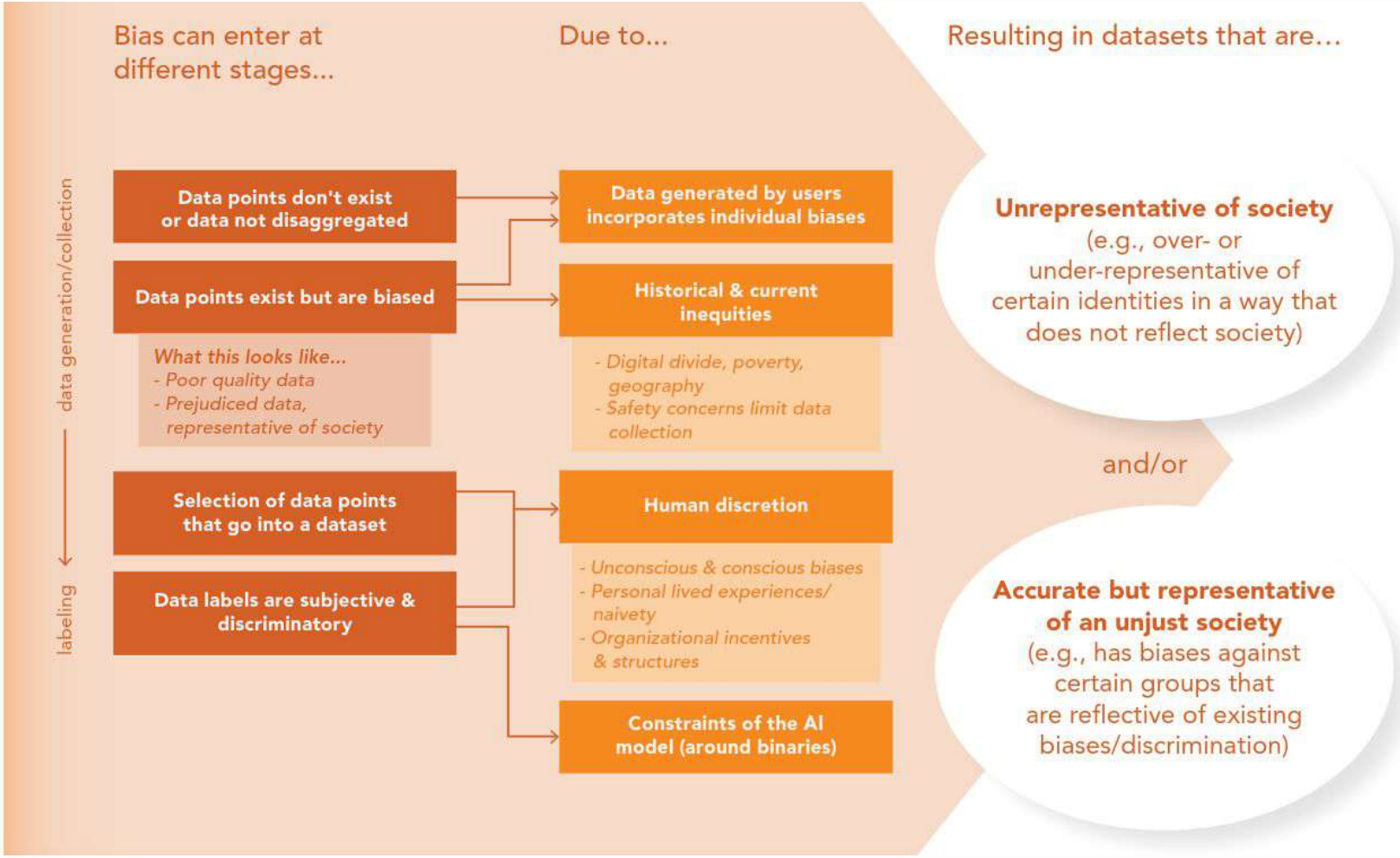
- When it comes to assigning gender labels, most data end up classified in terms of simplistic, binary female / male categories.

- When gender classification collapses gender in this way, it **reduces the potential for gender fluidity** and **self-held gender identity.**

1. Age distribution is skewed towards 25 years median.
2. Race or Ethnicity is skewed towards White Caucasian.
3. Race or Ethnicity is skewed against Indian.
4. There is a bias for Males over Females.

UTKFace dataset:

- **20,000** face images with annotations of age, gender, and ethnicity.
- **Race is categorized from zero to four**, and places people in one of five classes: White, Black, Asian, Indian, or “Others.”



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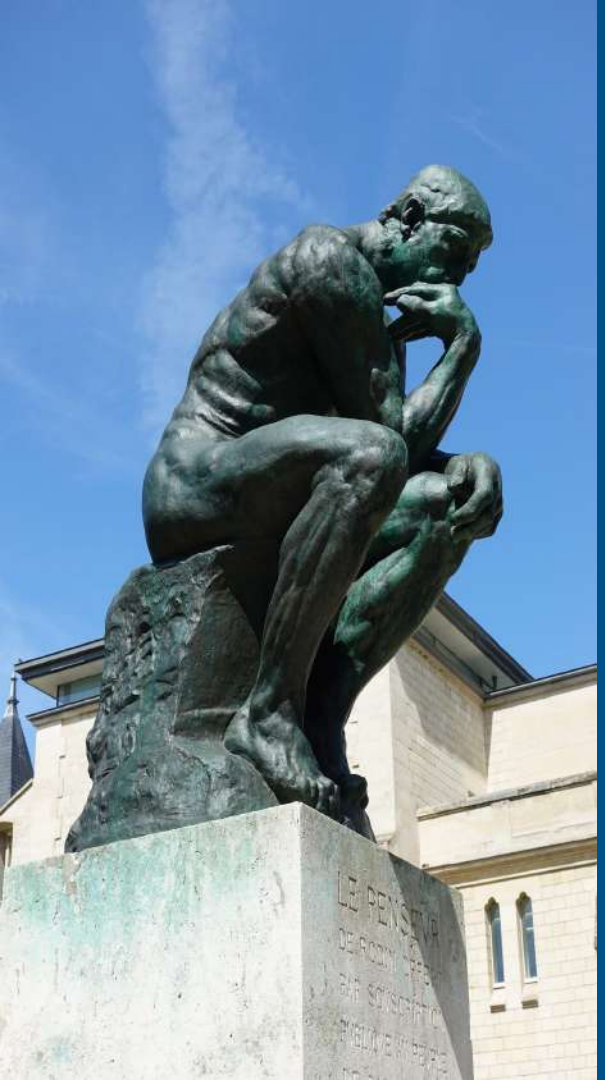
"Artificial intelligence reflects the values, priorities, and prejudices of its creators and annotators."



If we knew that an AI system was biased due to a biased training dataset, would you still trust its recommendations or decisions?

What biases can exist in a training dataset for an AI tool used in loan approval decisions in India, and how might these biases affect the loan approval decisions?